# Displacement and Infrastructure Provision: Evidence from the Interstate Highway System

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# Infrastructure construction displaces millions

- Infrastructure projects spur economic growth and development
  - Highways expand trade and raise output in U.S. cities (Duranton et al., '14; Herzog, '21)
- Construction of such projects is associated with the displacement of residents
  - Around 20M people were displaced in 2011 alone (World Bank, '18)
  - Displacement costs fall disproportionately on vulnerable populations (Penz et al., '11)
- We know little about the costs to **displaced** individuals and **nearby residents**

# The Interstate Highway System displaced over 1 million people




Overtown, Miami.




Source: Segregation by Design

The Interstate Highway System is the largest infrastructure project in U.S. history

- More than 4,500 km of highways built inside cities
- Estimated to have displaced more than 1M people (Schwartz, '75)
- No law requiring relocation aid for the first nine years of construction

# This paper: large non-pecuniary costs of Interstate construction

1. **Construction disproportionately affected the Black population**
   - Develop a new method to identify displaced people in the 1950 Census
   - Document that Black individuals were more likely to be displaced or live near highways
   - Results driven by local decisions rather than federal routing

2. **Negative long-run effects on displaced and near-highway residents**
   - Compare **displaced** and **adjacent** ($\leq$100m) to comparable unaffected neighbors (100–200m)
   - Displaced individuals die younger and relocate to worse neighborhoods in the city
   - Longevity also declines for those living near highways
   - Causes are the lack of relocation assistance and the erosion of housing wealth

# Contributions to the literature

1. **Long-run effects of displacement on individuals**
   - Existing work on war and natural disasters (Nakamura et al., '21; Deryugina et al., '18)
   - This paper: focus on policy-induced displacement

2. **Rising costs of infrastructure provision in the U.S.**
   - Literature centers on construction costs and local resistance (Liscow, '25; Brooks & Liscow, '23)
   - This paper: non-pecuniary costs borne by displaced individuals

3. **Consequences of the Interstate Highway System**
   - Costs & benefits for cities and neighborhoods (Baum-Snow, '07; Brinkman & Lin, '24; Weiwu, '25)
   - This paper: effects on people, not places

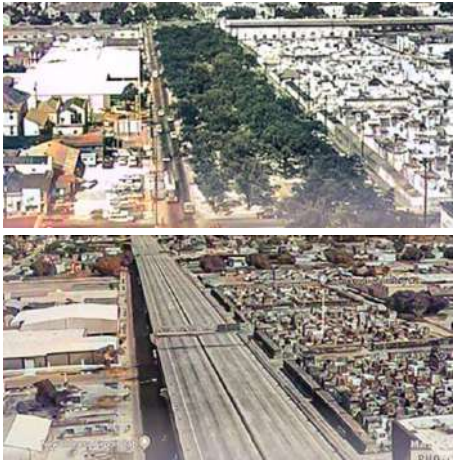
# The Interstate Highway System

# The 1956 Highway Act gave states full control over routing

## Federal-Aid Highway Act of 1956

- Disconnection between routing and funding
  - Federal government funded 90% of costs
  - States had full authority over routing
  - Federal engineers drafted routes for 100 cities (Yellow Book)
- More than 4,500 km built within cities
  - Displaced ∼1 million people (Schwartz, '75)

Tremé neighborhood, New Orleans



Before (top) and after (bottom)

# Officials used construction to carry out local agendas



Demolition of Overtown, Miami, for I–95



Source: Rothstein, ’17

“Some city officials expressed the view in the mid-1950s that the urban interstates would give them an opportunity to get rid of the local ‘n*****towns’ ”

Alfred Johnson, Exec. Director of AASHO — Schwartz, ’75

# Most construction occurred without relocation assistance

- No relocation assistance until 1965
  - Only homeowners were compensated
  - No support for moving or finding a new home
  - 15% secured housing through the program after relocation assistance was provided
    (Highway Research Board, '67)
- No systematic records of who was displaced

Black Bottom, Detroit



Before (top) and after (bottom)

# Identifying displaced individuals

# Novel method to identify who lived in destroyed homes

Identifying displaced individuals involves two steps:

1. Geocode the address information in the 1950 census
   - Use ArcGIS's 2023 geocoding engine ▸ Validation
   - Geocoded dwellings for all 1950 metro areas
   - Exact latitude and longitude for each household

Neighborhoods in Cleveland, OH



# Novel method to identify who lived in destroyed homes

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1. Geocode the address information in the 1950 census
   - Use ArcGIS's 2023 geocoding engine ▸ Validation
   - Geocoded dwellings for all 1950 metro areas
   - Exact latitude and longitude for each household
2. Highway locations from OpenStreetMap (2017)
   - Opening dates from PR511 (Baum-Snow, '07)
   - Focus on segments built between 1956 and 1970

Neighborhoods in Cleveland, OH



# Novel method to identify who lived in destroyed homes

**Note:** Residence observed in 1950, construction began in 1956

- Limited anticipation: data two years before Eisenhower's election
- No evidence of different underlying moving probabilities
- Estimates correspond to a lower bound

▸ Probability of moving out ▸ Bounding ▸ Correction ▸ Robustness ▸ Outmigration

## Benchmarking:

- **30,500** people displaced each year from 1956 to 1970
  → Highway Research Board: 33,070 displaced in 1965

Neighborhoods in Cleveland, OH

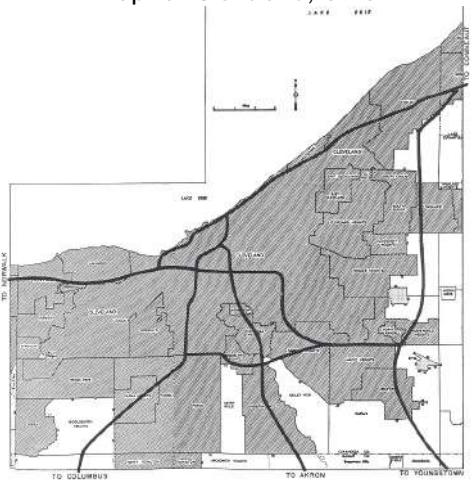


# Federal plans minimized costs

## Bureau of Public Roads Plans:

- Made in 1955 for 100 cities
- Also known as the “Yellow Book”
- Result of a cost minimization problem
- Usually aligned with primary roads
- No evidence of political motivations (Weiwu, ’25)

Map for Cleveland, Ohio



# Linking census records to administrative mortality records

Link the 1950 census to death records between 1988 and 2007

- To maximize matches:
  1. IPUMS MLP algorithm: $\approx$ 80% of links (Ruggles et al., '25)
  2. ABE-EI algorithm: $\approx$ 20% of links (Abramitzky et al., '25)
- Match rate of $\sim$ 26% in line with the literature (Abramitzky et al., '25; Ruggles et al., '25)
- Mortality records include two key proxies for economic wellbeing:
  1. **Location of the last residence:** characteristics of the last residence's neighborhood
  2. **Longevity:** most important determinant of lifetime wellbeing (Jones & Klenow, '16)

▸ Weights ▸ Prob. linkage ▸ Links

**Takeaways:**

Identify displaced individuals in the 1950 census

⇕

Administrative mortality records between 1988 and 2007

# Who was affected by construction?

# No systematic records of displaced people existed

- Historians argue highways were routed through low-income and Black neighborhoods
- This evidence comes from city case studies (Rose & Mohl, ’12; Rothstein, ’17; Archer, ’20)
- Use the geocoded dataset to empirically assess these claims

# Who lived where the highways were built?

$$y_i = \beta_0 \text{ Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{ Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i$$

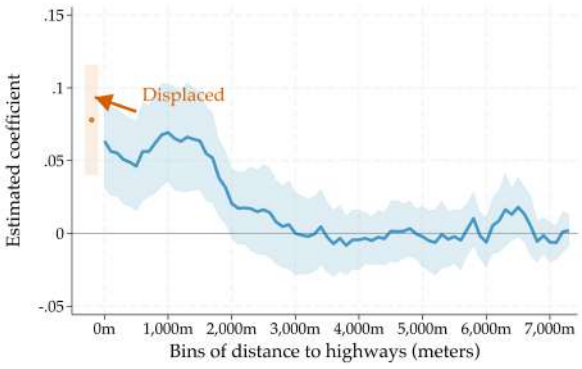
$i$: individual

- $y_i$: socioeconomic characteristic of person $i$ in 1950
- $\text{Displaced}_i$, $\text{Bin}_i^k$: distance bin to future highway (comparison group: 7.4–7.5 km)
- $\beta_0$, $\beta_k$: difference in characteristics relative to those “far” from the highway

# Highway construction disproportionately affected Black communities

$$y_i = \beta_0 \text{ Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{ Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i$$

Black




People living closer to highways were:

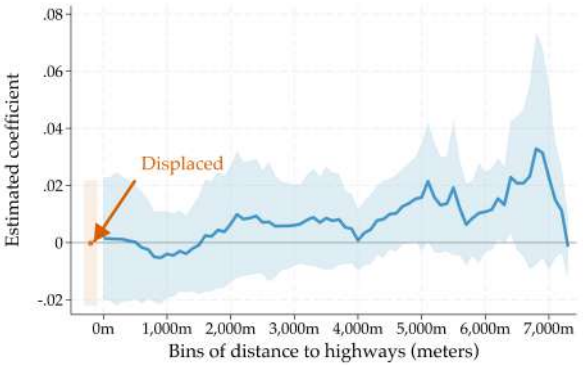
- More likely Black $\sim$ 7 pp

▸ Additional outcomes ▸ No controls ▸ Census tract analysis

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Foreign born




People living closer to highways were:

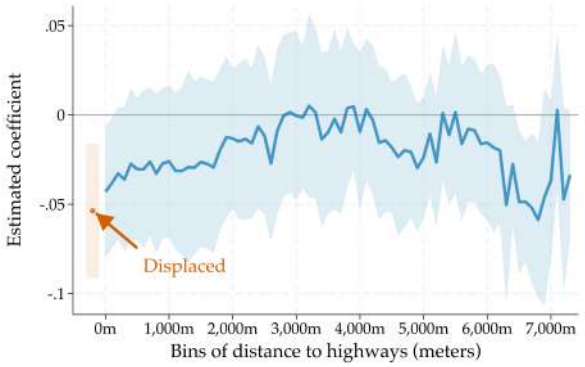
- More likely Black $\sim$ 7 pp
- Not more likely foreign born

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$$y_i = \beta_0 \text{ Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{ Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i$$

Income wage (log)




People living closer to highways were:

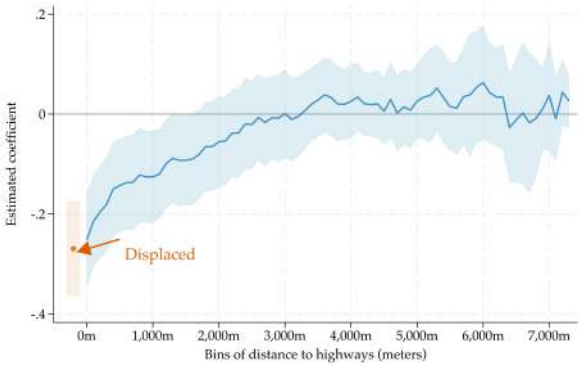
- More likely Black $\sim$ 7 pp
- Not more likely foreign born
- Earning lower wages $\sim$ 0.05 log points

▸ Additional outcomes ▸ No controls ▸ Census tract analysis

# Highway construction disproportionately affected Black communities

$$y_i = \beta_0 \text{ Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{ Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i$$

Home value (log)




People living closer to highways were:

- More likely Black $\sim$ 7 pp
- Not more likely foreign born
- Earning lower wages $\sim$ 0.05 log points
- In cheaper homes $\sim$ 0.25 log points
  - Imputed from the 1940 census for those in the same home

▸ Additional outcomes ▸ No controls ▸ Census tract analysis

# Was displacement of Black communities inevitable?

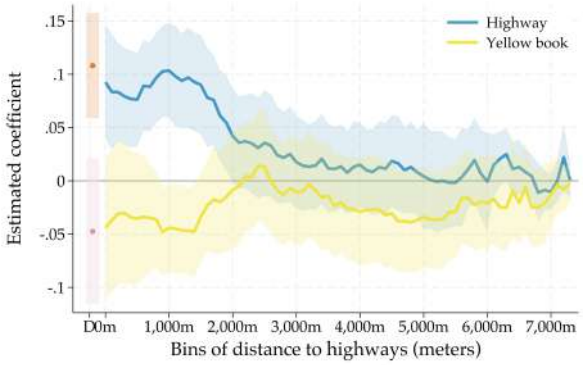
- Black families sorted into downtown areas with more jobs (Boustan, '10)
- Highways were designed to connect city centers

  → Any counterfactual network might also have passed through Black neighborhoods?
- I compare characteristics along the Yellow Book plans to the built network

# Built highways targeted Black communities more than federal plans did

$$y_i = \beta_0 \, \text{Disp}_i + \sum_{k \in \mathcal{K}} \beta_k \, \text{Bin}_i^k + \delta_0 \, \text{Disp_YB}_i + \sum_{k \in \mathcal{K}} \delta_k \, \text{Bin_YB}_i^k + \mathbf{X}_i' \, \Gamma + \epsilon_i$$

Black




Planned routes were:

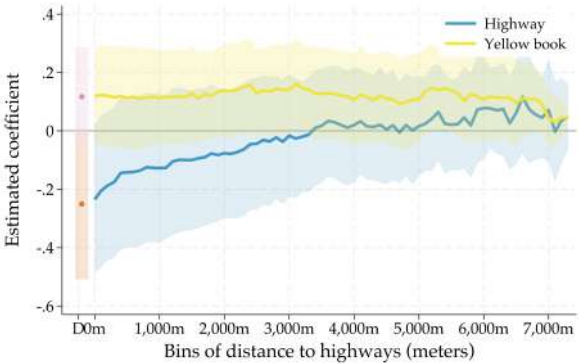
- Less likely to affect Black residents

▸ Coefficient test

# Built highways targeted Black communities more than federal plans did

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Home value (log)




Planned routes were:

- Less likely to affect Black residents
- Through areas with higher home values

▸ Coefficient test

## Takeaways:

Black Americans were disproportionately affected by highway construction

The impact was due to state and local planners' choices, not the federal plans

# Long-run effects of displacement

# Comparing the displaced to comparable unaffected neighbors

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

$i$: individual

**Research design:**

- Compare to observationally similar unaffected neighbors living farther away
- Trade-off:
  1. Too close may be affected by highway construction
  2. Too far away may be differently selected

$\rightarrow$ Empirical exercise guides the design

# Selecting a comparison group

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

$i$: individual

**Exercise:** How long-run outcomes evolve with distance to the highway:

1. Negative spillovers within 100 meters of the highway
2. No evidence of spillovers beyond 100 meters

▸ Exercise

# Empirical strategy

Long-run effects for those displaced by construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

**Strategy:** Unaffected neighbors

- Two treatments:
  - **Displaced** by highways to study long-run costs

Neighborhoods in Cleveland, OH



# Empirical strategy

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**Strategy:** Unaffected neighbors

- Two treatments:
  - **Displaced** by highways to study long-run costs
  - **Adjacent** ($\leq$ 100m) to study localized spillovers

Neighborhoods in Cleveland, OH



# Empirical strategy

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Strategy: Unaffected neighbors

- Two treatments:
  - **Displaced** by highways to study long-run costs
  - **Adjacent** ($\leq$ 100m) to study localized spillovers
- Compare to those **living between 100–200m**
  - Distance is independent of potential outcomes

Neighborhoods in Cleveland, OH



# Empirical strategy

Long-run effects for those displaced by construction:

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Strategy: Unaffected neighbors

- Two treatments:
  - **Displaced** by highways to study long-run costs
  - **Adjacent** ($\leq$ 100m) to study localized spillovers
- Compare to those **living between 100–200m**
  - Distance is independent of potential outcomes
- $\mathbf{X}_i$: race, gender, birth-year FE, head's occ. score, metro FE

▸ Wealth

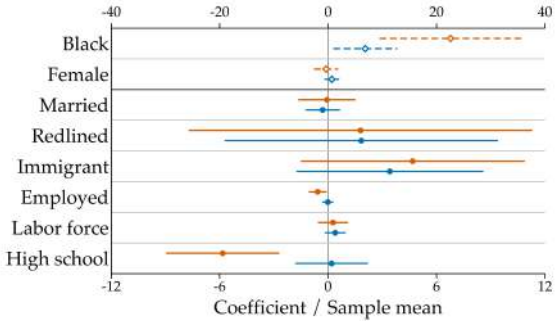
Neighborhoods in Cleveland, OH

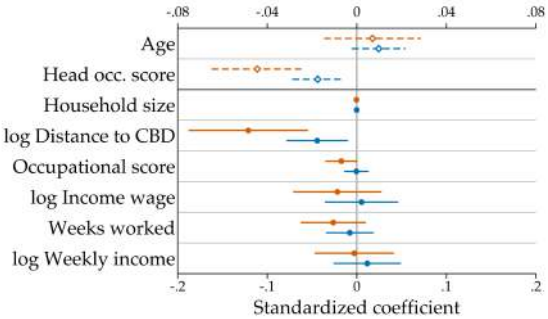


# Balance test

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$




▸ Pre-trends

# Alternative strategies

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

$i$: individual

Two complementary strategies to deal with possible threats

- Matching based on observable characteristics to individuals living $\geq$ 2,000m away
  - Addresses spillovers onto the 100–200m control group ▸ Balance ▸ Trends
- Compare to those living under unbuilt Yellow Book segments
  - Addresses selection on unobservables near built routes ▸ Balance ▸ Trends

▸ Maps

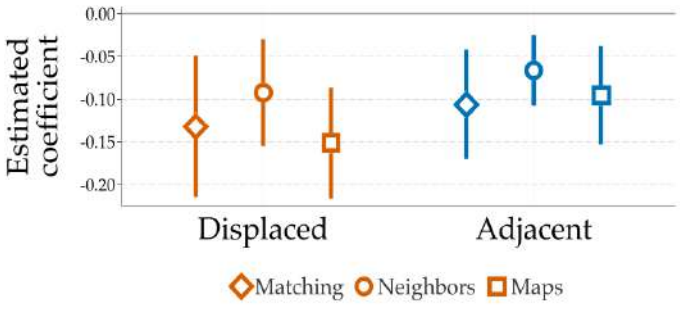
# Mortality effects

# Displaced individuals die approximately one month earlier

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Death age




Displaced and adjacent **die** younger

- Effect is $\sim$ 1 month for displaced
- 1/3 of the homeless–housed gap
  (Meyer et al., '23)


▸ Predicted mortality ▸ Placebo ▸ Corrected ▸ Farther control group ▸ Robustness

# Displaced individuals die approximately one month earlier

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Survive past 75 years old




Displaced and adjacent **die** younger

- Effect is $\sim$ 1 month for displaced
- 1/3 of the homeless–housed gap (Meyer et al., '23)
- No effect on surviving past 75

▸ Predicted mortality ▸ Placebo ▸ Corrected ▸ Farther control group ▸ Robustness

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Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Survive past 80 years old




Displaced and adjacent **die** younger

- Effect is $\sim$ 1 month for displaced
- 1/3 of the homeless–housed gap (Meyer et al., '23)
- No effect on surviving past 75
- Lower prob. surviving past 80


▸ Predicted mortality ▸ Placebo ▸ Corrected ▸ Farther control group ▸ Robustness

# Spatial mobility after displacement

# Use last recorded residence to study spatial mobility

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

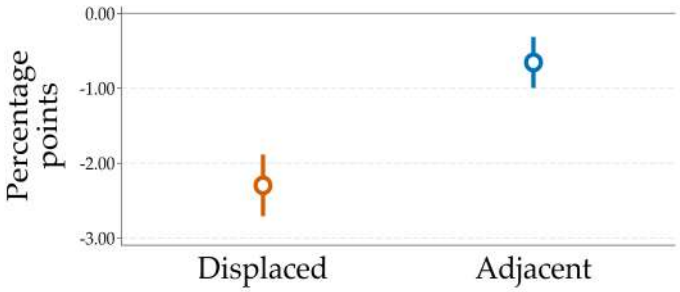
- I use the residence at death to study spatial mobility
  - Geocode the 9-digit ZIP code to identify the neighborhood of residence

# Displaced leave neighborhoods but stay in the same city

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Same neighborhood




Displaced and adjacent **leave** their neighborhood

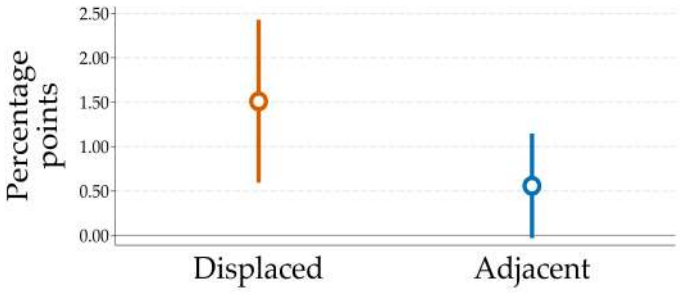
▸ All strategies ▸ Placebo ▸ Corrected ▸ Farther control group ▸ Robustness

# Displaced leave neighborhoods but stay in the same city

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Same city




Displaced and adjacent leave their neighborhood

Relocate **within** the same city

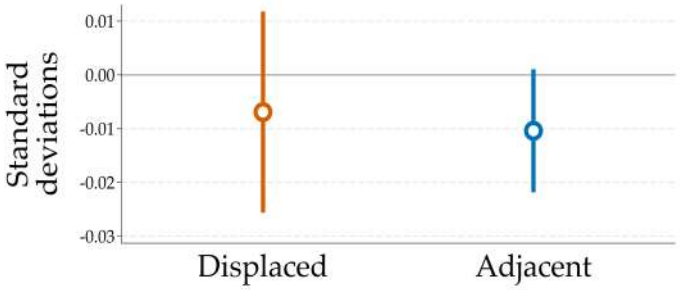
All strategies | Placebo | Corrected | Farther control group | Robustness

# Displaced leave neighborhoods but stay in the same city

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Distance to the CBD (log)




Displaced and adjacent leave their neighborhood

Relocate within the same city

Remain in the city center

All strategies | Placebo | Corrected | Farther control group | Robustness

# Characteristics of the neighborhoods they relocate to

# Characteristics of the neighborhoods they move to

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \, \Gamma + \epsilon_i$$

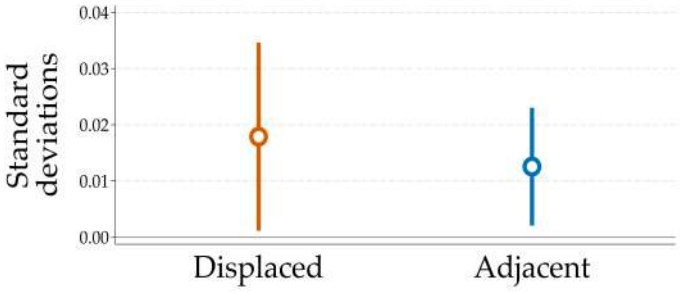
- Use characteristics of the census tract of residence in 2000
- Variables are standardized to have mean zero and standard deviation of one

# Characteristics of the neighborhoods they move to

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Unemployment share




Displaced & adj. live in tracts with:

**Higher** unemployment share

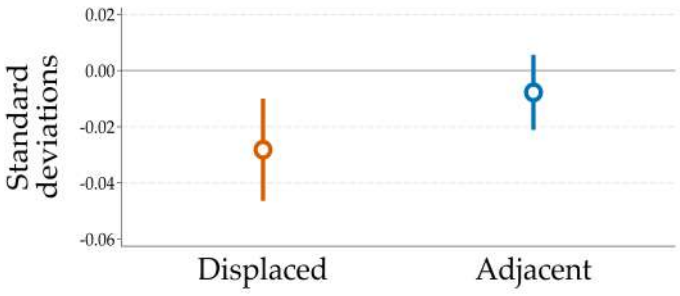
All strategies

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Dep. var.: College share




Displaced & adj. live in tracts with:

Higher unemployment share

**Lower** college share

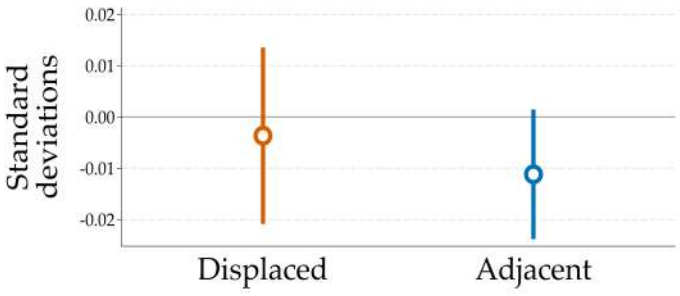
▸ All strategies

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$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Homeownership share




Displaced & adj. live in tracts with:

- Higher unemployment share
- Lower college share

No clear differences in:

- Homeownership share

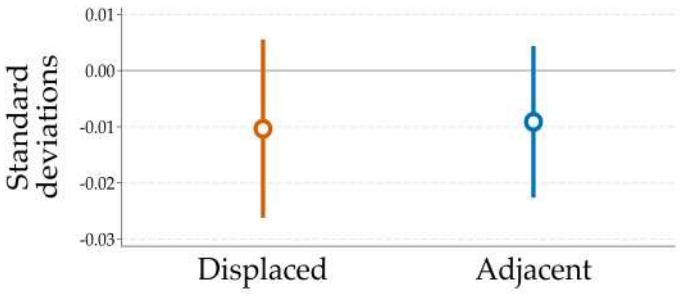
▸ All strategies

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Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Median income (log)




Displaced & adj. live in tracts with:

- Higher unemployment share
- Lower college share
- No clear differences in:
  - Homeownership share
  - Median income

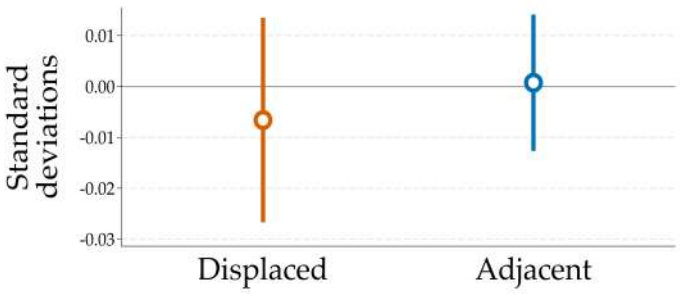
All strategies

# Characteristics of the neighborhoods they move to

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Median home value (log)




Displaced & adj. live in tracts with:

- Higher unemployment share
- Lower college share

No clear differences in:

- Homeownership share
- Median income
- Median home value

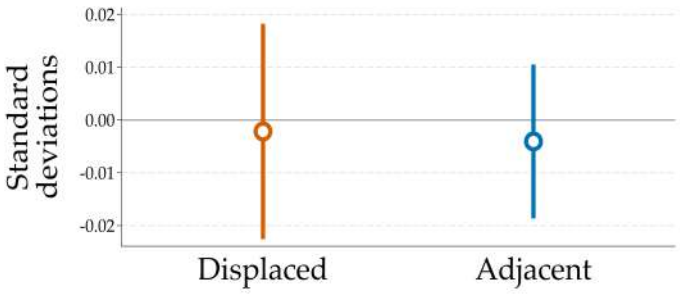
▸ All strategies

# Characteristics of the neighborhoods they move to

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Median rent (log)




Displaced & adj. live in tracts with:

- Higher unemployment share
- Lower college share

No clear differences in:

- Homeownership share
- Median income
- Median home value
- Median rent

▸ All strategies

# Displaced individuals relocated to worse neighborhoods

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

- I combine all these characteristics into a **quality index**

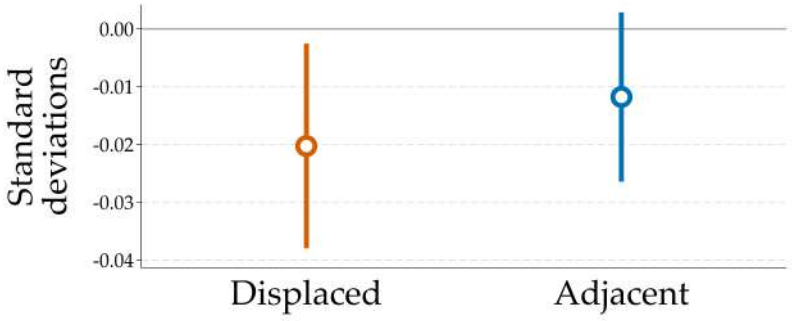
→ First principal component

# Displaced individuals relocated to worse neighborhoods

Long-run effects for those displaced by highway construction:

$$y_i = \beta_1 \text{ Displaced}_i + \beta_2 \text{ Adjacent}_i + \mathbf{X}_i' \Gamma + \epsilon_i$$

Dep. var.: Quality index




▸ All strategies

# Were Black and white residents affected differently?

# Black residents were more exposed to displacement

- Displaced residents were $\sim$ 7 pp more likely to be Black
- Black families relocated within a segregated housing market (Cutler et al., '99)

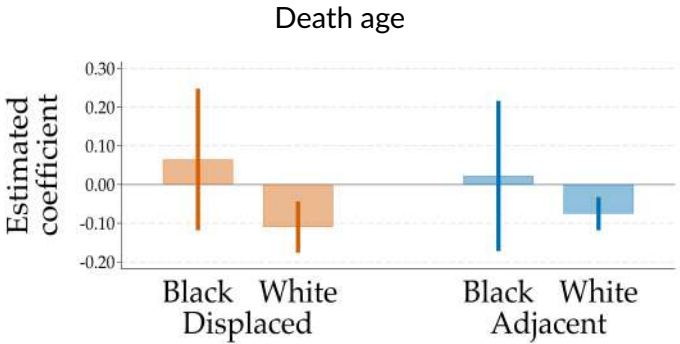
  $\rightarrow$ Did Black residents bear larger costs from displacement?

- I estimate the effects separately for Black and white residents

# Effects come entirely from white residents

Long-run effects by race:

$$y_i = \beta_1^B \text{ Displaced}_i^B + \beta_1^W \text{ Displaced}_i^W + \beta_2^B \text{ Adjacent}_i^B + \beta_2^W \text{ Adjacent}_i^W + \mathbf{X}_i' \Gamma + \epsilon_i$$




Results driven by white residents

- Displaced die $\sim$ 6 weeks earlier
- Adjacent die $\sim$ 4 weeks earlier
- No effect for Black residents

Not due to pre-existing differences

Largest losses outside redlined areas

▸ Mobility ▸ Reweighting ▸ Redlining

**Takeaways:**

- Displaced individuals die younger and relocate to worse neighborhoods in the same city
- Adjacent individuals also die younger and move out
- Results driven by white residents

# What explains the mortality effects?

a. Neighborhoods and labor markets
b. Erosion of housing wealth
c. Cost of relocation

# What explains the mortality effects?

a. **Neighborhoods and labor markets**

b. Erosion of housing wealth

c. Cost of relocation

# Neighborhoods and labor markets do not explain the effects

1. Destinations were not worse when individuals moved ▸ Details
2. Differences in origin neighborhoods do not explain the results ▸ Details
3. Local labor markets were similar during working years ▸ Details
4. Air pollution and distance to the center barely differ ▸ Details

# What explains the mortality effects?

a. Neighborhoods and labor markets ✗
b. Erosion of housing wealth
c. Cost of relocation

# What explains the mortality effects?

a. Neighborhoods and labor markets ✗

**b. Erosion of housing wealth**

c. Cost of relocation

# Highways eroded the value of nearby homes

"[Construction] left the ground and adjacent buildings in shadow and economically depressed"

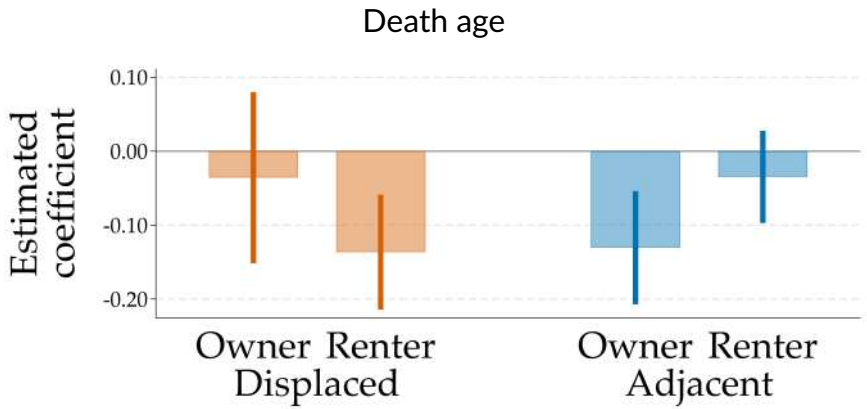
Lewis, '13

- Housing was the main asset of most American households (Akbar et al., '25)
- Highways reduced the value of homes
  - Eminent domain paid below market value (Rose & Mohl, '12)
  - Highways depressed nearby property values (Brinkman & Lin, '24)

→ Effects should concentrate among homeowners*

* Tenure from linked 1940–1950 censuses

# Housing wealth explains the effect for adjacent residents




Erosion of housing wealth explains the results for those **adjacent**

Redlining Race

# What explains the mortality effects?

a. Neighborhoods and labor markets ✗

b. Erosion of housing wealth ✓ but only for those **adjacent**

c. Cost of relocation

# What explains the mortality effects?

a. Neighborhoods and labor markets ✗

b. Erosion of housing wealth ✓

c. **Cost of relocation**

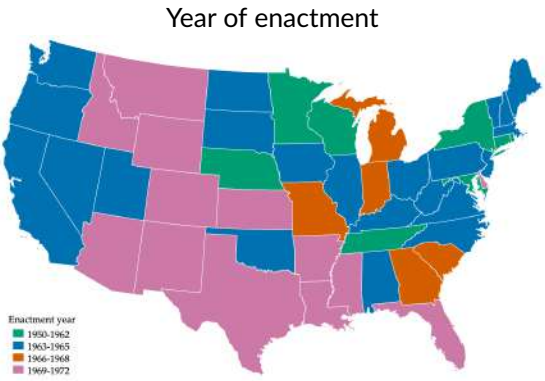
# Condemnations occurred with little notice and no help

“They knocked on the door, gave her a dollar and said ‘The house is ours.’ ”

Fred Salvucci, on his grandmother’s eviction — Lewis, ’13

# States adopted relocation payments at different times

- Displaced received no help to move
  - Only 11 states paid by 1963 (HRB, ’63)
- I collect each state’s year of enactment
  - Most states adopted in 1962–1969



# Comparing those displaced before and after their state's law

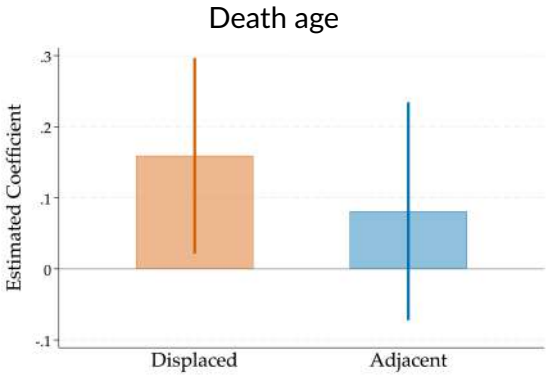
$$y_i = \gamma_1 \text{Disp}_i + \gamma_2 \text{Disp}_i \times R_i + \delta_1 \text{Adj}_i + \delta_2 \text{Adj}_i \times R_i + \lambda_{t(i)} + \mathbf{X}_i' \Gamma + \epsilon_i$$

- $R_i$: construction began after the state enacted relocation assistance
- $\gamma_2$: effect of assistance on those displaced
- $\delta_2$: placebo, since adjacent residents were not entitled to payments
- $\lambda_{t(i)}$: year-of-construction fixed effects

Balance

# Relocation assistance offset the mortality cost of displacement

Effects before and after the state enacted relocation assistance:




Coefficients for post-enactment for displaced and adjacent residents

label Pre-law: displaced **die** younger

- $\sim$ 2 months earlier

Post-law: the effect disappears

- Gain of $\sim$ 2 months

No significant change for adjacent

No effects found for mobility and neighborhood quality

# Assistance likely worked through housing security

"If we can't find housing, we can't build highways"

Lowell Bridwell, FHWA director, 1968 — Rose & Mohl, '12

- Laws required replacement housing before homes could be acquired (Rose & Mohl, '12)
  - Often met with public housing (Kushner & Werner, '76)
- Displaced movers were 3 times as likely to enter public housing ▸ AHS
  - Descriptive evidence from movers in the 1973–1974 AHS
- Explains why assistance did not change where they moved

# Conclusions

# Displacement had lasting costs, but payments could offset them

- Highways disproportionately affected Black communities
- Displaced die younger and relocate to worse areas in the same city
- Adjacent residents also die younger and move out
- Relocation payments offset nearly all mortality effects

Thank you! You can contact me at: pablo.valenzuela-casasempere@yale.edu

# Appendix

# Geocoding benchmark

Using modern locators to geocode historical addresses may lead to measurement error

1. Changes in house numbering
2. Changes in street names

Use historical street grids for 9 cities to benchmark the accuracy of modern locators

Albany & Buffalo, NY; Cincinnati, Cleveland & Dayton, OH; Louisville, KY; Spokane, WA; Springfield, MA; St. Louis, MO

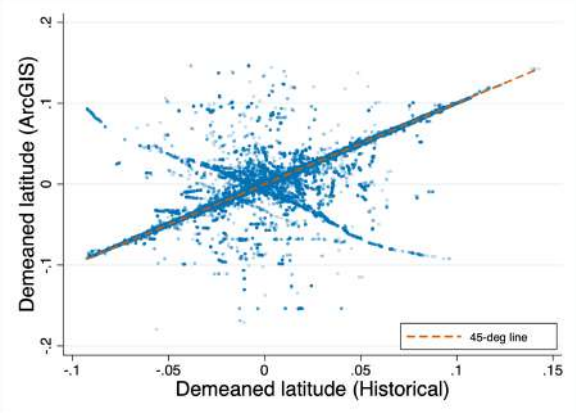
## Results:

- Treatment assignment is highly correlated across geocoders Go
- Correlation between normalized coordinates from modern and historical locators Go
- CDF of distance between the two coordinates Go
- Directional bias in the geocoding error Go

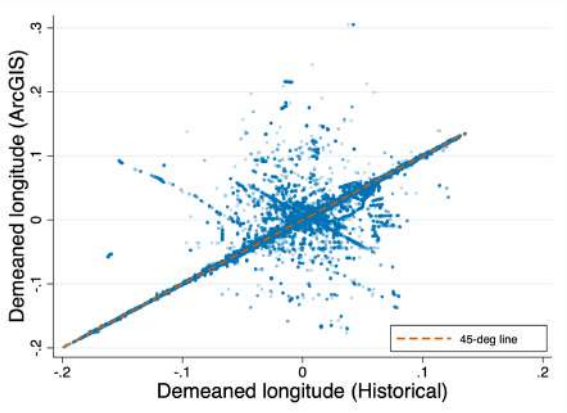
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# Geocoding benchmark: Correlation of coordinates

Latitude

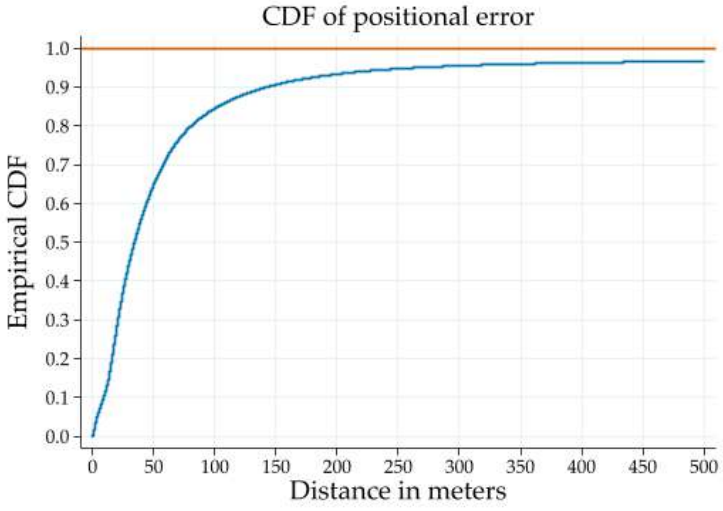



Longitude



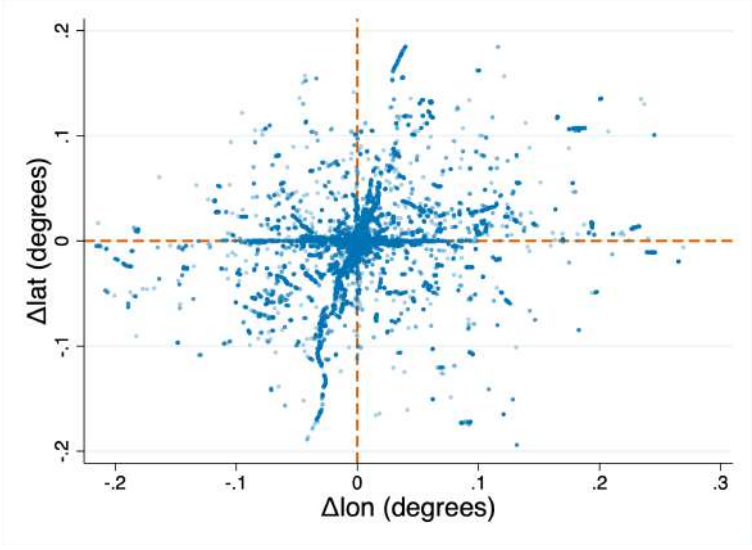

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# Share of addresses with both coordinates within a given distance




Back

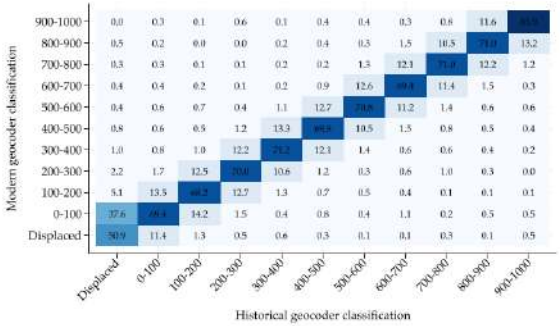
# Directional bias in the error of the geocoding



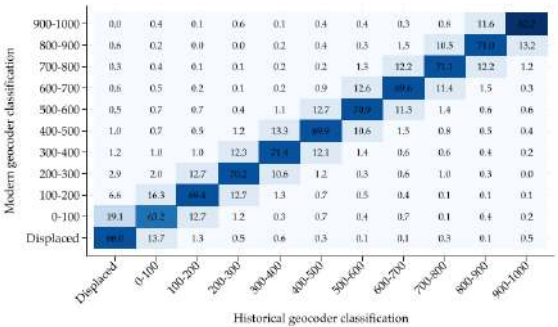

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# Treatment assignment is highly correlated across geocoders

All




Excluding 25m




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# Bounding the effect of displacement

I estimate the following equation:

$$y_i = \beta \cdot D_i^{50} + \epsilon_i$$

where $D_i^{50}$ is a dummy equal to one if $i$ lived in 1950 in a house destroyed by construction.

The *true* effect is given by:

$$y_i = b \cdot D_i^{56} + e_i$$

where $D_i^{56}$ is a dummy equal to one if $i$ was actually displaced by construction.

The estimated effect corresponds to:

$$\hat{\beta} \xrightarrow{p} \mathbb{E}(y_i|D_i^{50}=1) - \mathbb{E}(y_i|D_i^{50}=0) = b \cdot \Big(\mathbb{E}(D_i^{56}|D_i^{50}=1) - \mathbb{E}(D_i^{56}|D_i^{50}=0)\Big)$$

$$\implies b \in \left[\hat{\beta}, \frac{\hat{\beta}}{\mathbb{P}(D_i^{56}=1|D_i^{50}=1) - \mathbb{P}(D_i^{56}=1|D_i^{50}=0)}\right] \approx [\hat{\beta}, 3 \cdot \hat{\beta}]$$

Back

# Displaced were not more likely to move out before construction

## Intuition:

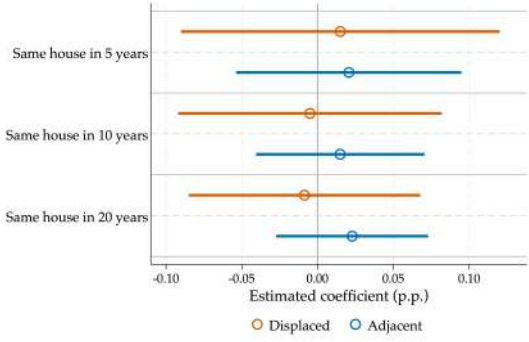
- Displaced may have higher mobility
- Use socioeconomic and demographic characteristics to predict mobility

## Predicted probability:

- Probab. of staying in the same home in the 1960 census (5, 10, and 20 years)
- Covs: age, race, gender, labor outcomes
- Out-of-sample prediction in 1950 census
- Control group: 100–200m from a highway

→ Coefficients are small and not significant

Probability of staying in the same house




Back

# Correcting for moves before construction

## Problem:

- We observe residence in 1950 ($D_i^{50}$)
- $S_i = 1$ if $i$ stayed until construction $\rightarrow$ *true* displacement is $D_i^{56} = D_i^{50} \cdot S_i$
- *True* effect $b$ is given by $D_i^{56}$, but we only observe $D_i^{50}$
- Estimated effect is a lower bound: $\beta = b \cdot \Pr(S_i = 1 \mid D_i^{50} = 1) \leq b$

## Correction:

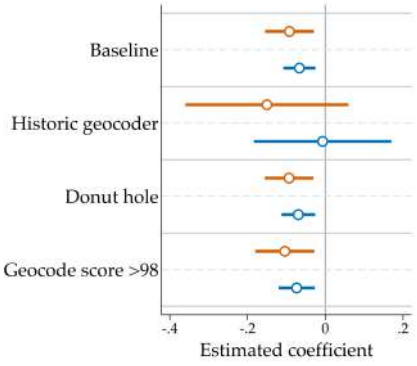
- I predict who stayed from observable characteristics $W_i$
- $s_i$ is the predicted probability of staying Details
- Replace $D_i^{50}$ with the expected true exposure: $\tilde{D}_i = \mathbb{E}(D_i^{56} \mid D_i^{50}, W_i) = D_i^{50} \cdot s_i$
- Regressing $y_i$ on the corrected exposure $\tilde{D}_i = D_i^{50} \cdot s_i$ recovers $b$

$\rightarrow$ Corrected effects are 2 to 2.7 times the baseline estimates

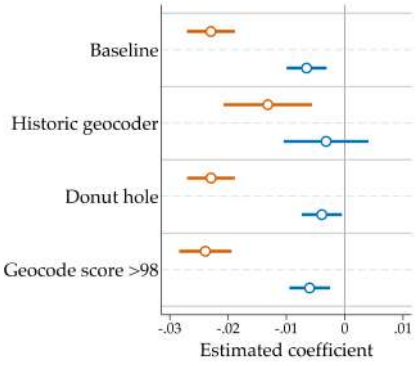
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# Geocoding robustness checks

Death age



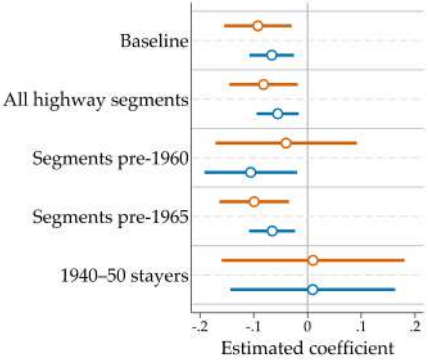

Same neighborhood



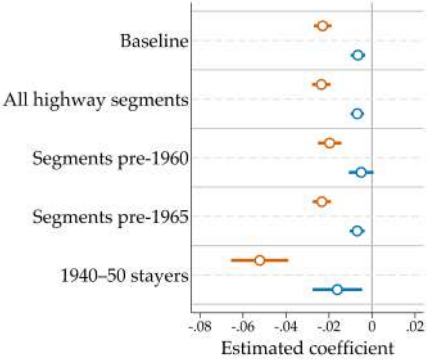

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# Samples less exposed to moves before construction

Death age



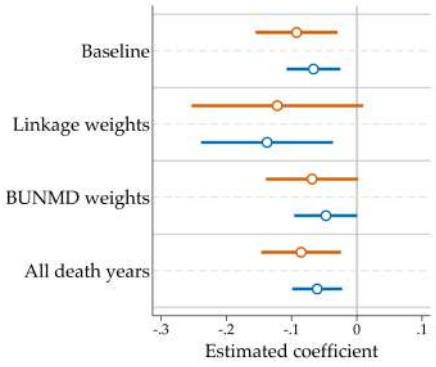

Same neighborhood



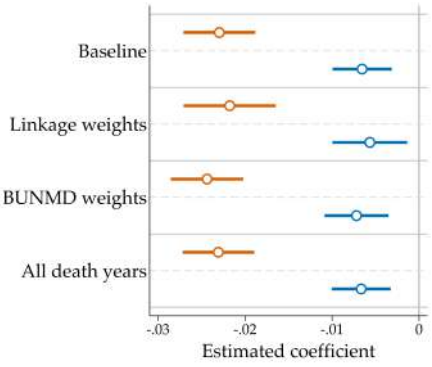

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# Robustness to weighting by probability of linkage

Death age



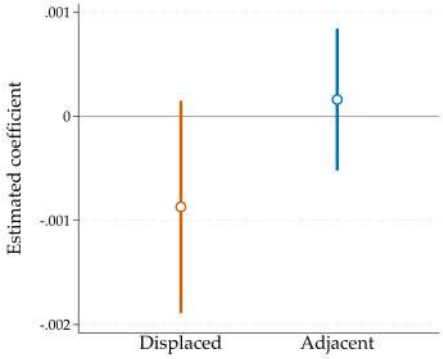

Same neighborhood




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# Prob. linkage to mortality records by treatment status

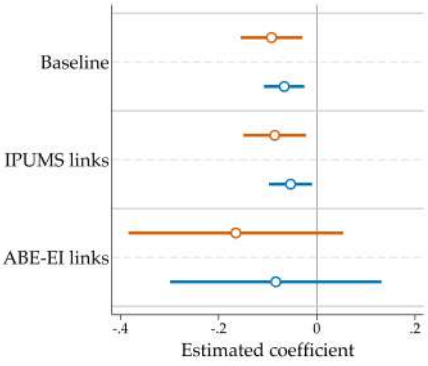
Linked to mortality records



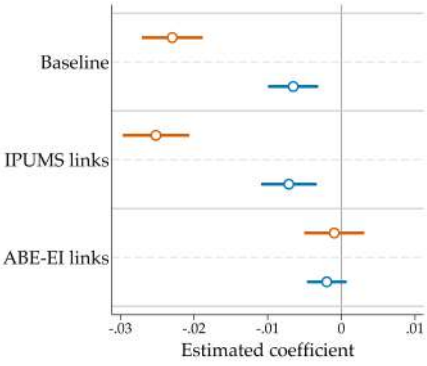

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# Different linking methods to mortality records

Death age




Same neighborhood

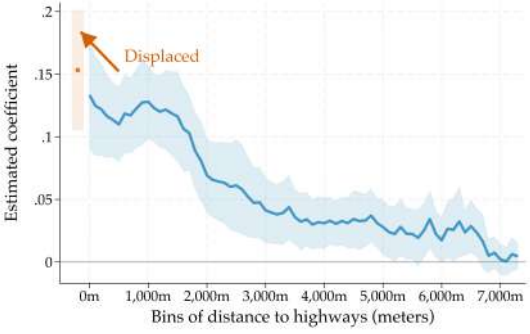



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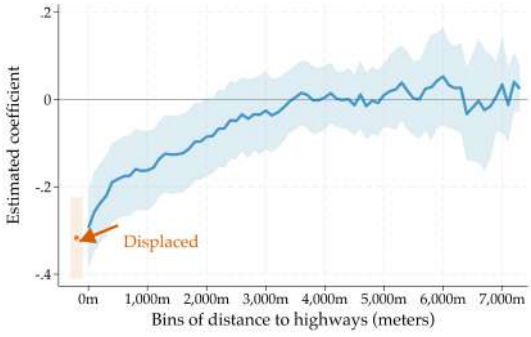
# Gradients without controlling for distance to the city center

$$y_i = \beta_0 \ \text{Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \ \text{Bin}_i^k + \lambda_c + \epsilon_i$$

Black




Home value (log)

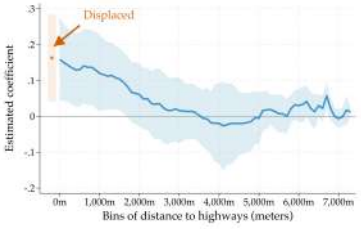



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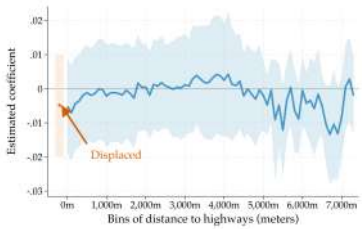
# Redlined, employment, and rents gradient

$$y_i = \beta_0 \text{ Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{ Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i$$

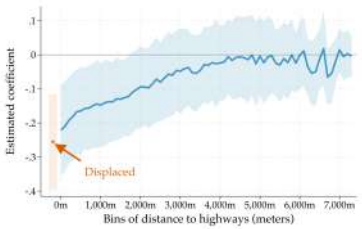
Redlined




Employment




Rent (log)

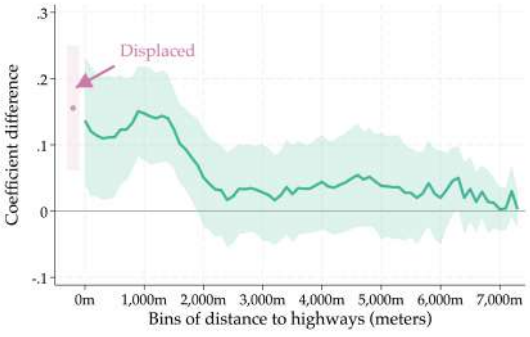



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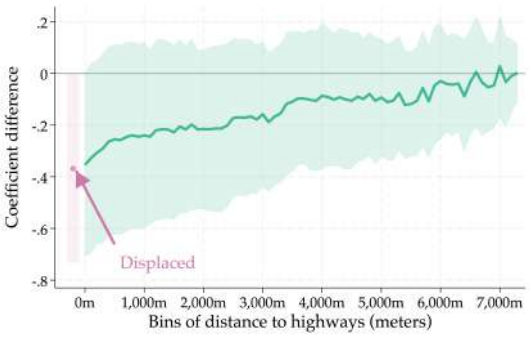
# Coefficient test

$$H_0 : \beta_k = \delta_k \qquad H_1 : \beta_k \neq \delta_k$$

Black




Home value (log)




Back

# Where are highways built?

Probability of construction as a function of the tract's characteristics in 1950

$$y_n = \beta_1 \, BlackShare_n + \beta_2 \log \mathbf{P}_n + \beta_3 \log DistCBD_n + \beta_4 \, Plan_n + \Gamma' \, \mathbf{X}_n + \epsilon_n$$

$n$: census tract

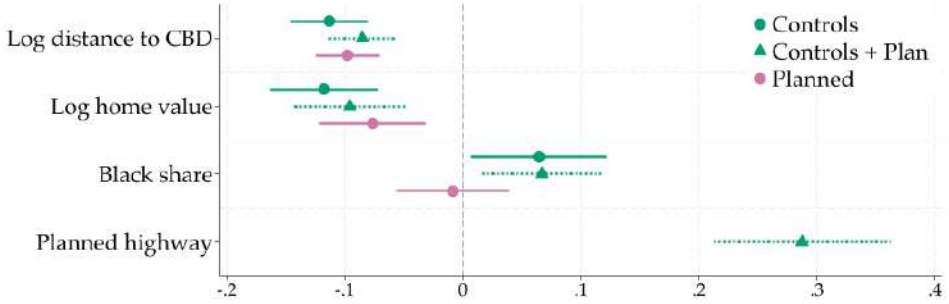
- $y_n$: indicator if a highway was built through the tract
- $BlackShare_n$: share of the city's Black population living in the tract
- $\mathbf{P}_n$: proxy for the median land price in the tract (home value and rent)
- $DistCBD_n$: (log) distance to the city's Central Business District
- $Plan_n$: indicator of planned highways in the tract

Back

# Where are highways built?

Probability of construction as a function of the tract's characteristics in 1950

$$y_n = \beta_1 \, BlackShare_n + \beta_2 \log \mathbf{P}_n + \beta_3 \log DistCBD_n + \beta_4 \, Plan_n + \Gamma' \, \mathbf{X}_n + \epsilon_n$$




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# Testing for spatial spillovers

The dataset allows me to test for spatial spillovers:

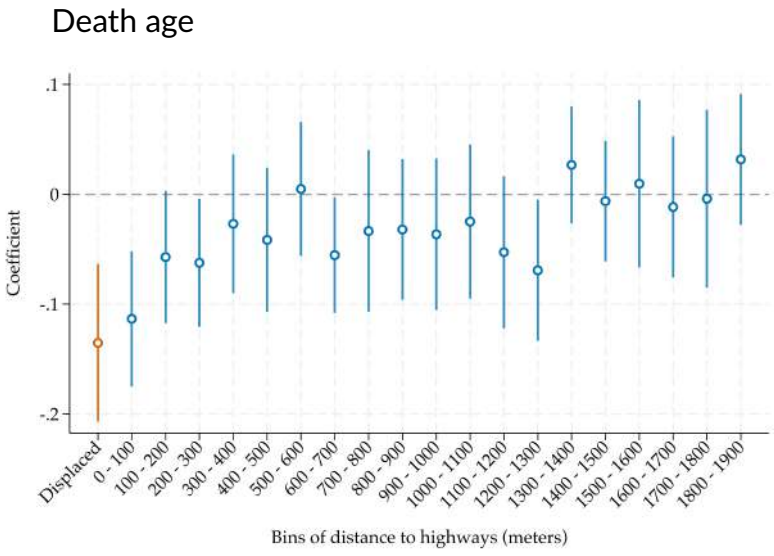
$$y_i^{2000} = \beta_0 \ \text{Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \ \text{Bin}_i^k + \mathbf{X}_i' \ \Gamma + \ \epsilon_i$$

$i$: individual

- Sample: 1950 census linked to mortality records, within 2 km of highways
- $y_i^{2000}$: death age and spatial mobility across neighborhoods and cities
- $\beta_0$, $\beta_k$: difference relative to those at 1,900–2,000m
- $\mathbf{X}_i$: gender, race, birth-year FE, head's occupational score, metro FE

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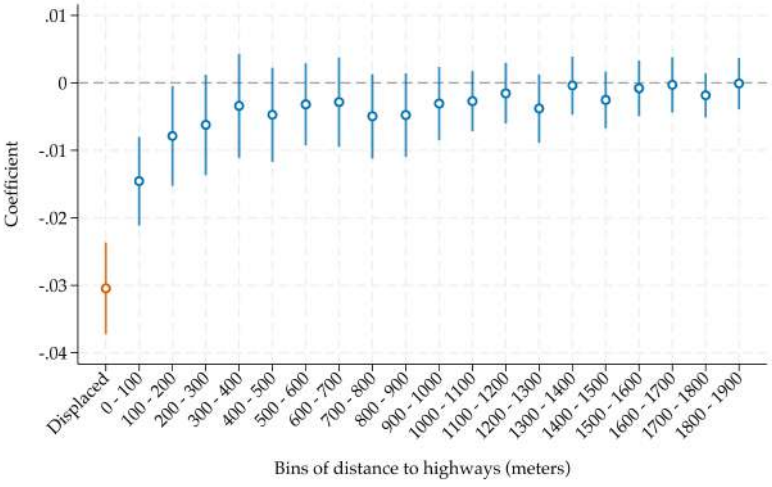
# Displaced and nearby residents die younger; effects vanish after 100m




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# Adjacent residents also leave their neighborhoods

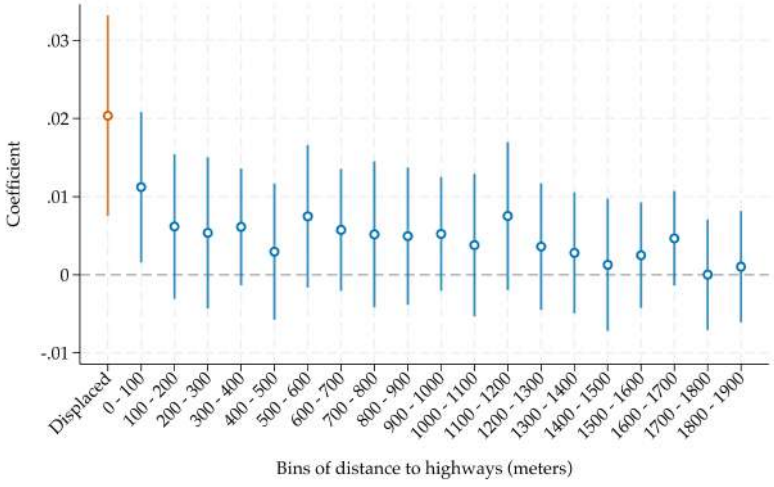
Same neighborhood




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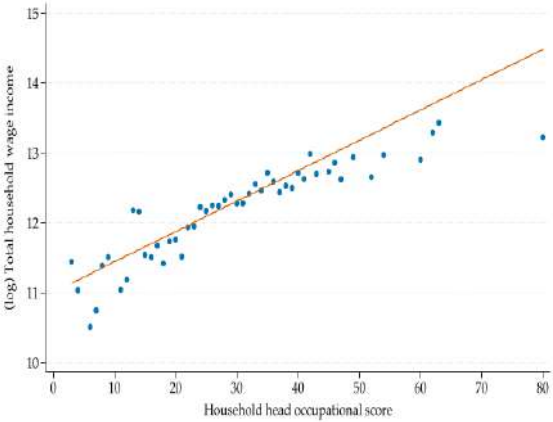
# Spillovers do not push people out of the city

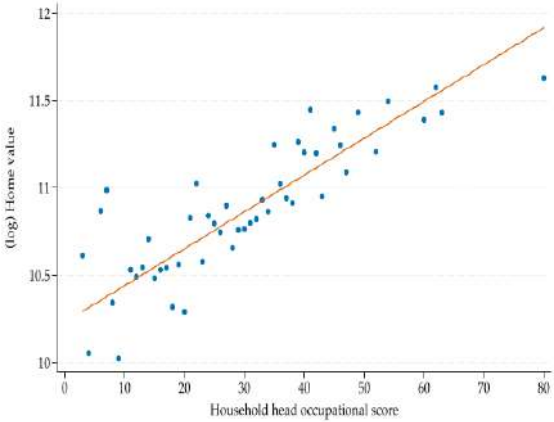
Same city in 1950 and at death




Back

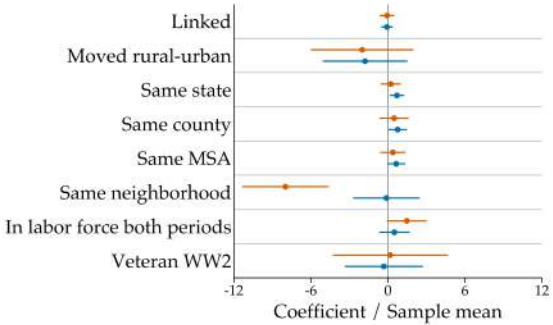
# Head's occupational score as a proxy for wealth

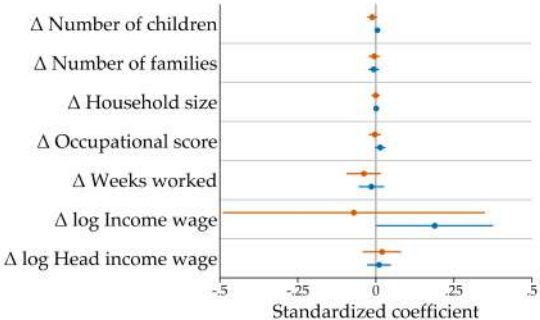



Back

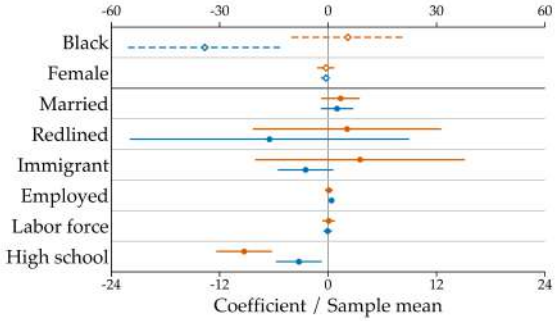
# Neighbors: No pre-trends between 1940 and 1950

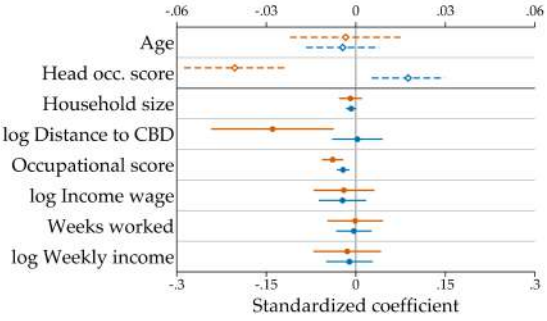



Back

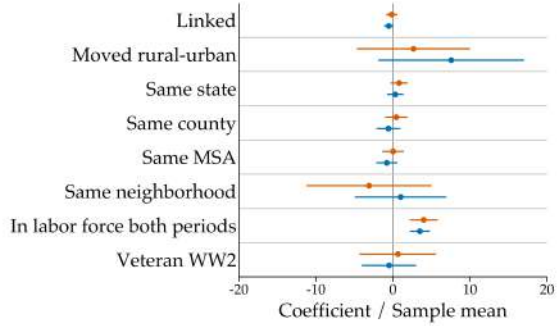
# Matching: Balance test

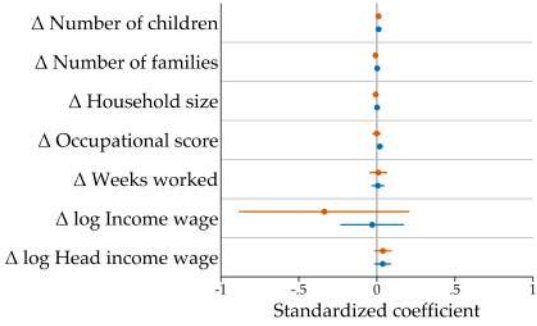



Back

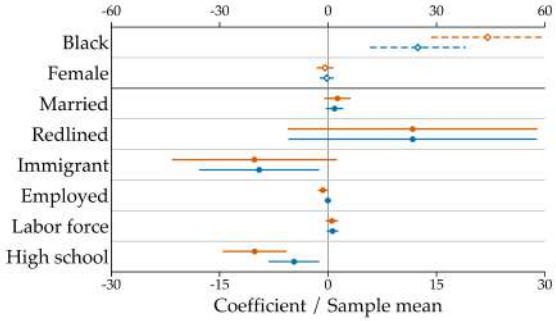
# Matching: No pre-trends between 1940 and 1950

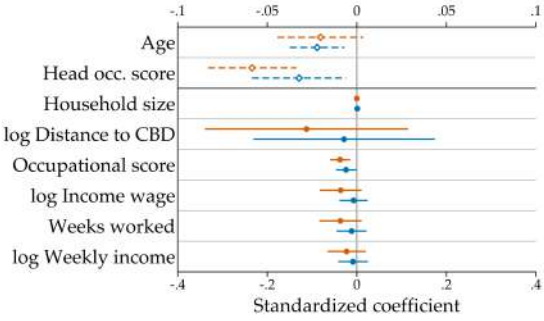



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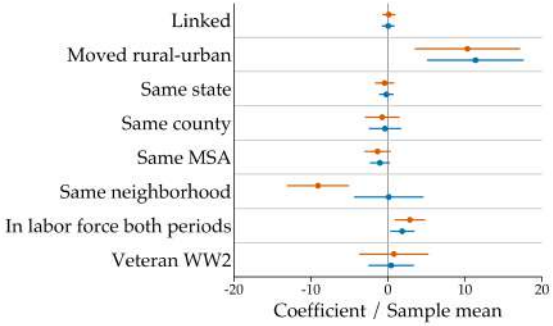
# Yellow Book: Balance test

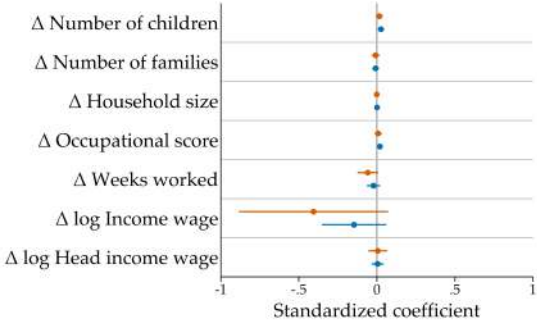



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# Yellow Book: No pre-trends between 1940 and 1950




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# Maps of complementary strategies

Matching

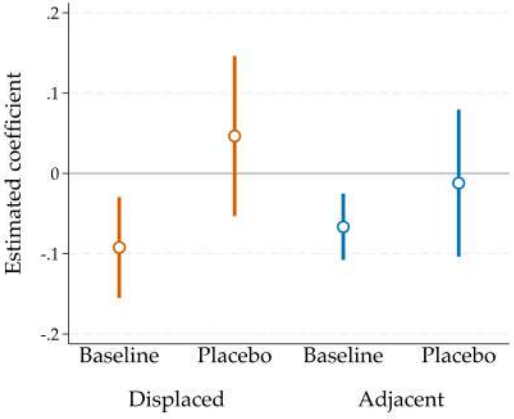



Yellow Book

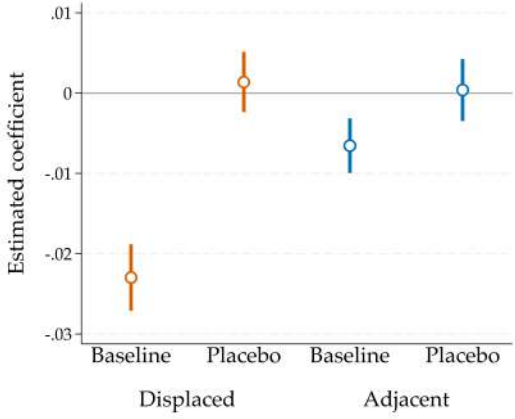


# Placebo using Yellow Book maps

Death age

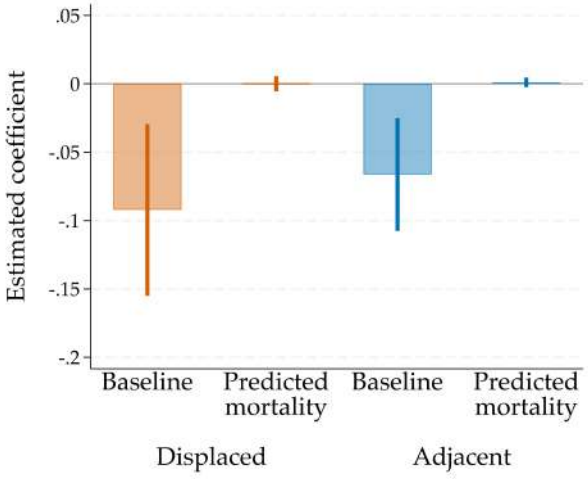



Same neighborhood



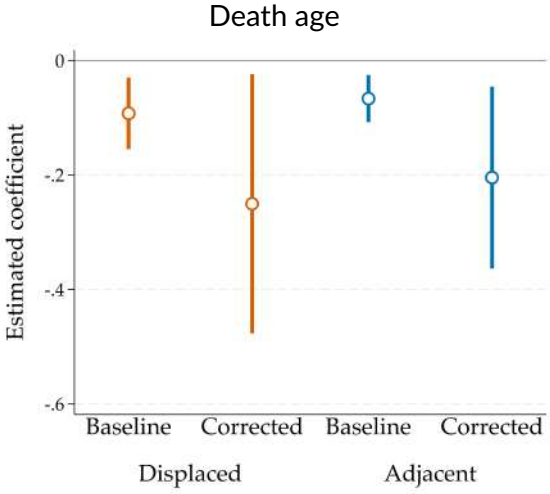

◂ Mortality ◂ Mobility

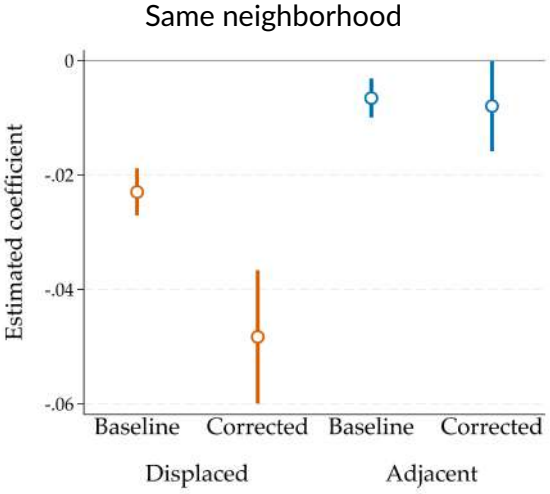
# Mortality predicted using socioeconomic characteristics




Mortality

# Results correcting for probability of staying

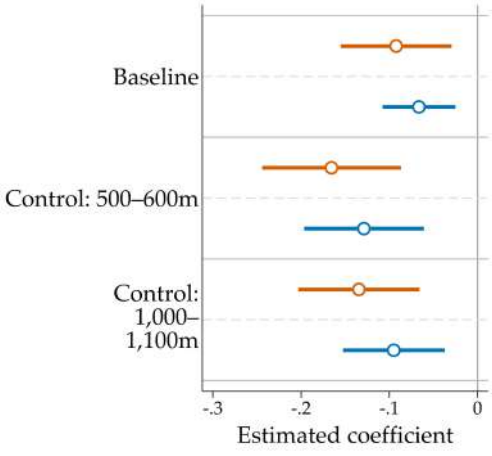


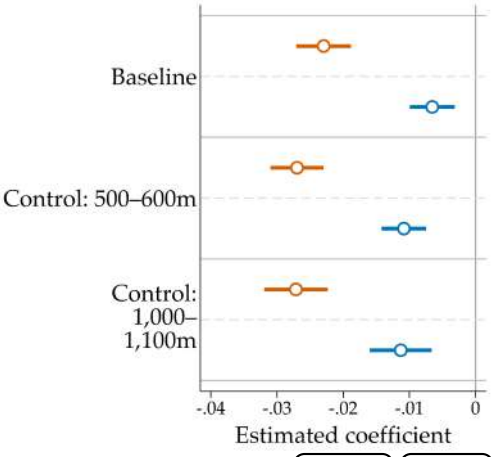

◂ Mortality ◂ Mobility

# Comparison group at farther distances

Death age



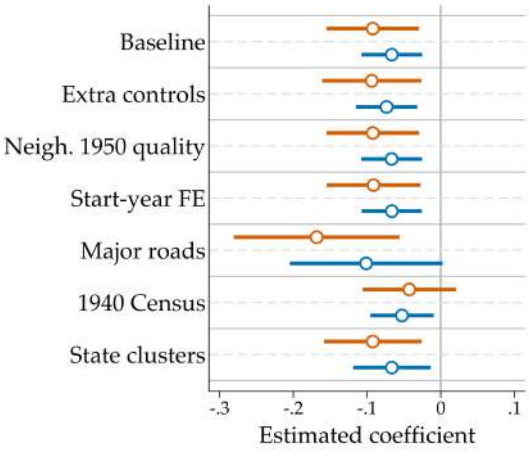

Same neighborhood



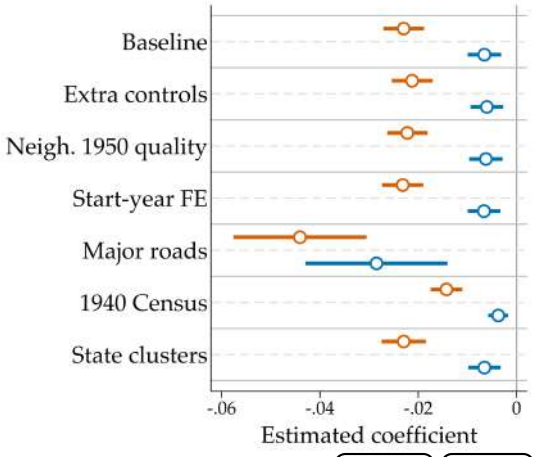

◂ Mortality ◂ Mobility

# Additional robustness checks

Death age



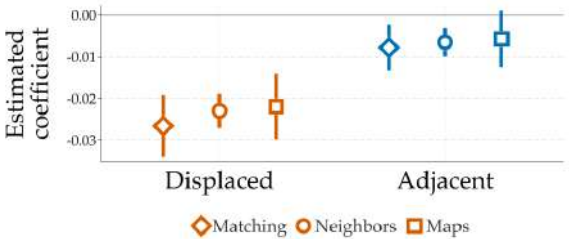

Same neighborhood




◂ Mortality ◂ Mobility

# Displaced leave their neighborhood under all three strategies

Same neighborhood




Same city



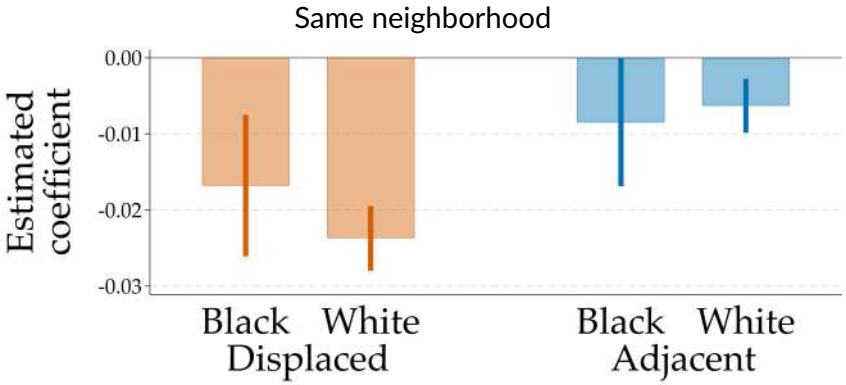

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# Displaced move to worse neighborhoods under all three strategies



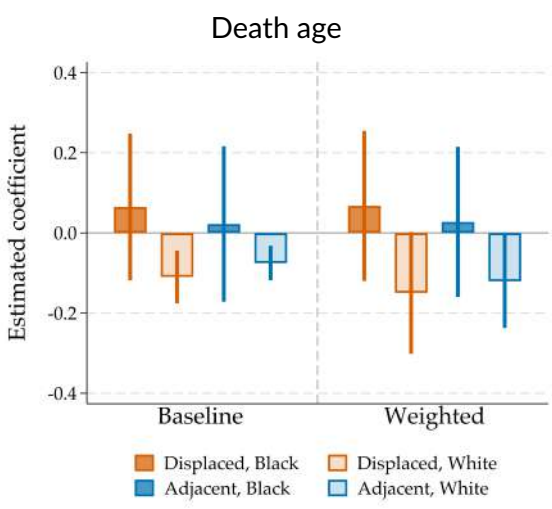

◂ Characteristics ◂ Quality index

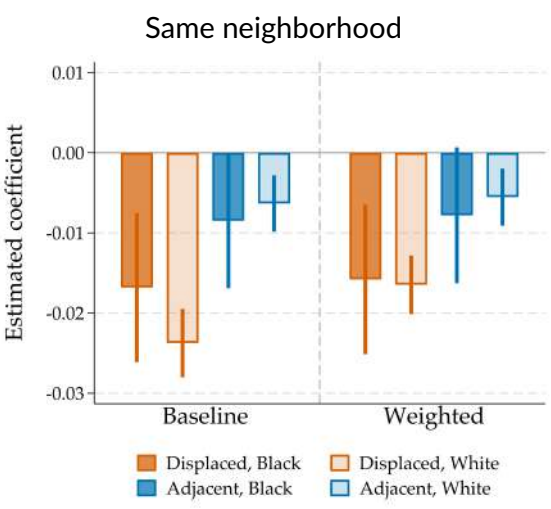
# Black and white residents both leave their neighborhoods




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# Racial gap persists after reweighting on 1950 characteristics

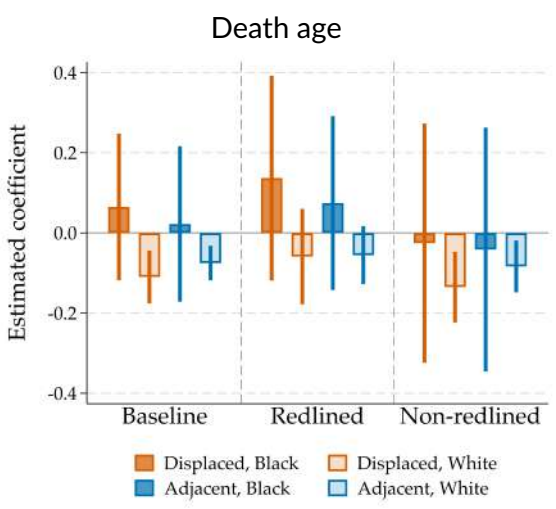


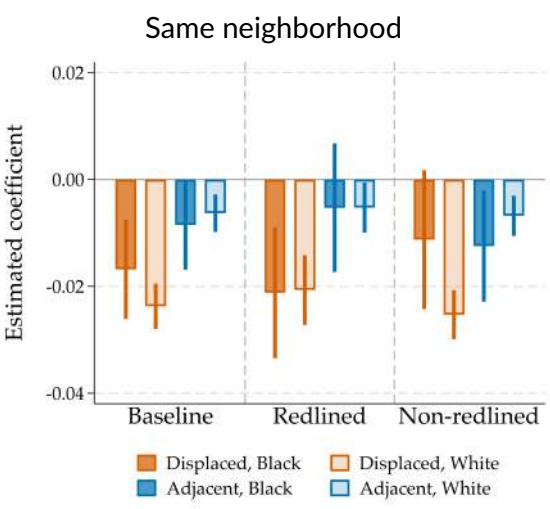




White residents reweighted to Black residents on metro, sex, head's occ. score, birth cohort, and 1940 tenure

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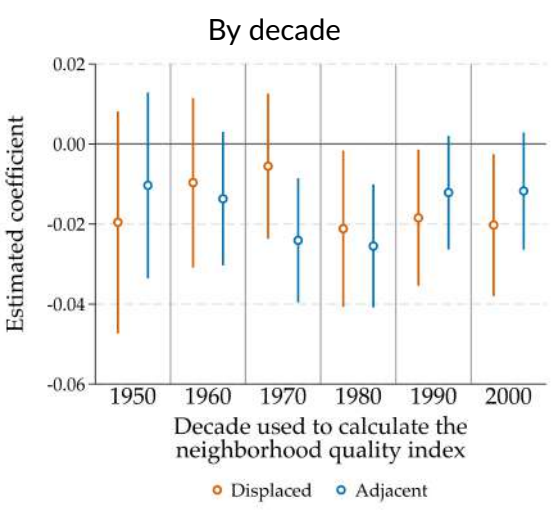
# Effects by race and redlined status of the 1950 neighborhood

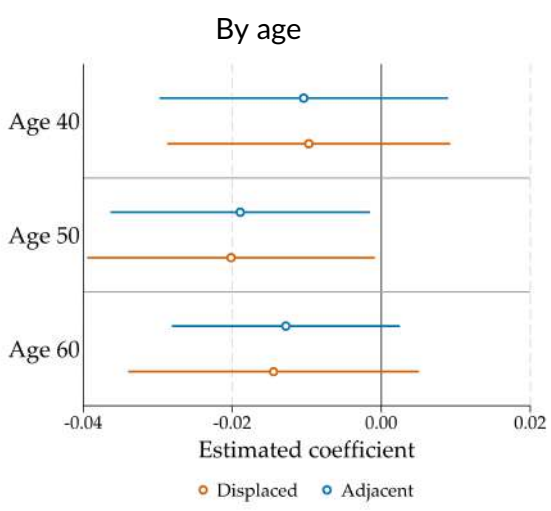



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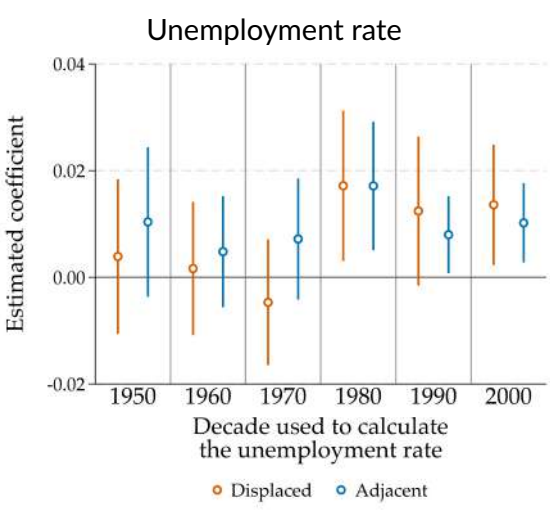
# Neighborhood quality falls only decades after the move

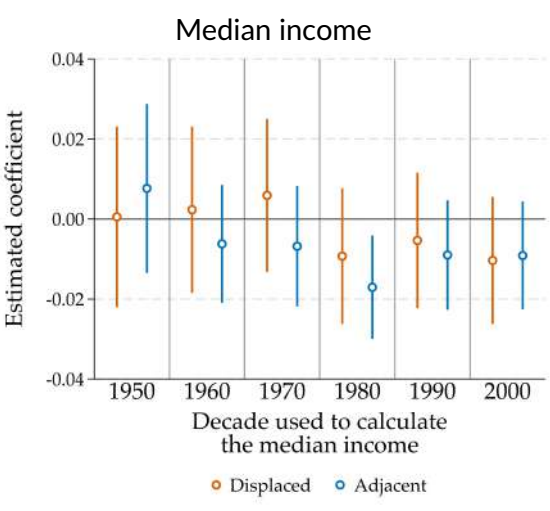



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# Local labor markets were similar during working years

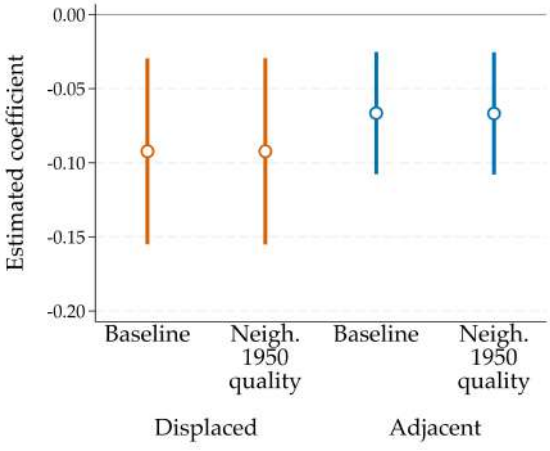


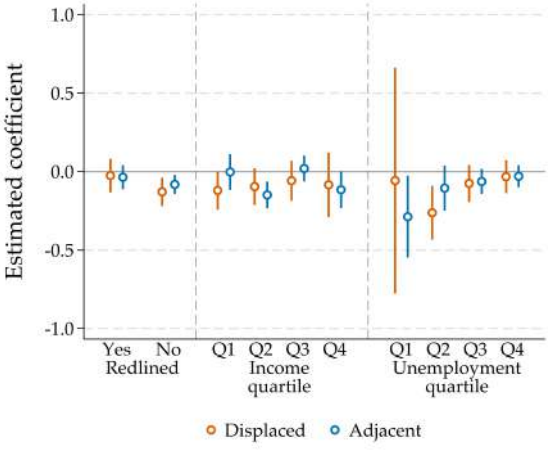

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# The 1950 neighborhood does not drive the results

## Controlling for 1950 quality




## By 1950 neighborhood




Dep. var.: death age

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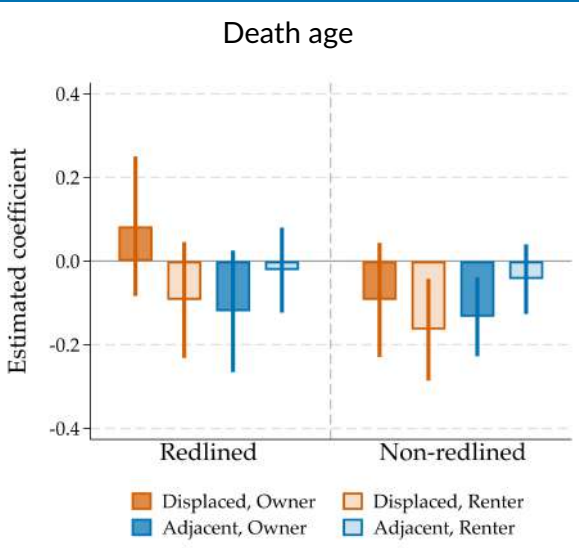
# Pollution and distance to the city center barely differ

| | $PM_{2.5}$ (SD) | log Distance to CBD |
|---|---|---|
| | (1) | (2) |
| Displaced | 0.014[c] | -0.007 |
| | (0.008) | (0.009) |
| Adjacent | 0.005 | -0.010[c] |
| | (0.005) | (0.006) |
| Observations | 121,124 | 121,322 |
| R-squared (adjusted) | 0.016 | 0.151 |

Neighborhood of residence at death. $PM_{2.5}$: 2000–2005 average, standardized. [a] 1%, [b] 5%, [c] 10%.

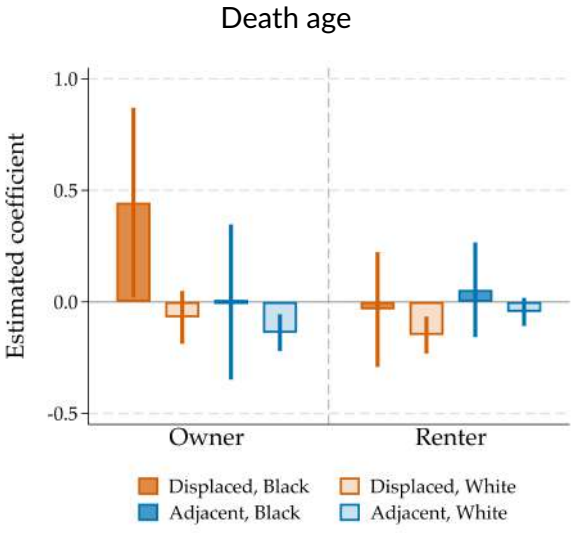
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# Effects by tenure and redlining




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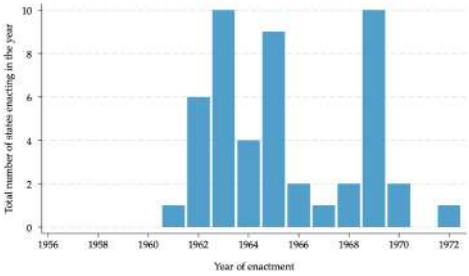
# Effects by race and tenure




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# Timing of enactment across states

## States enacting each year



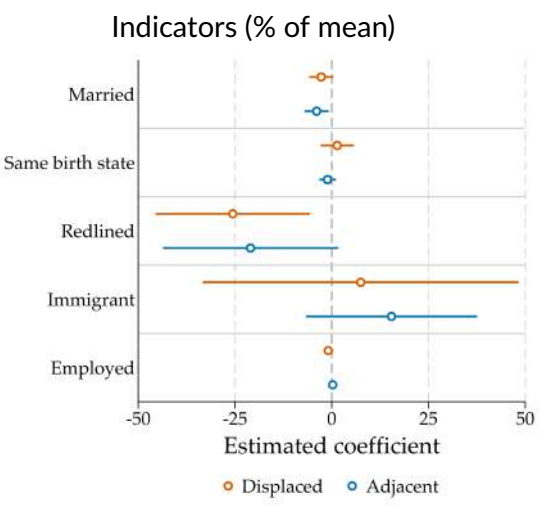

## Correlates of the year of enactment

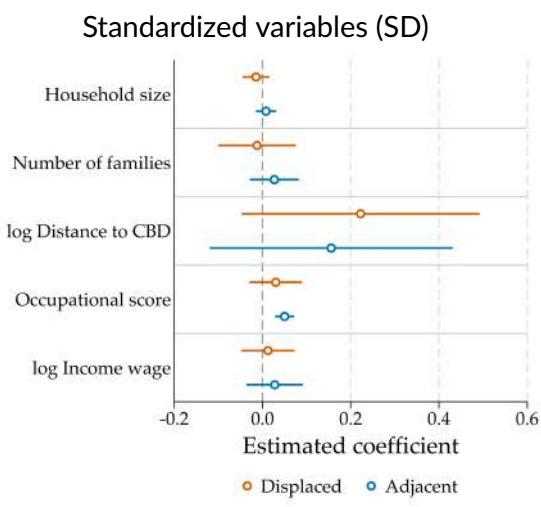
| | log GDP per capita (1963) | log Total Population (1960) | log White Population (1960) | log Non-White Population (1960) |
|---|---|---|---|---|
| Year of enactment | -2.258[c] | -0.308 | -0.430 | 0.240 |
| | (1.312) | (0.381) | (0.376) | (0.199) |
| Observations | 48 | 48 | 48 | 48 |
| R-squared | 0.036 | 0.012 | 0.024 | 0.025 |

Robust standard errors. [a] 1%, [b] 5%, [c] 10%.

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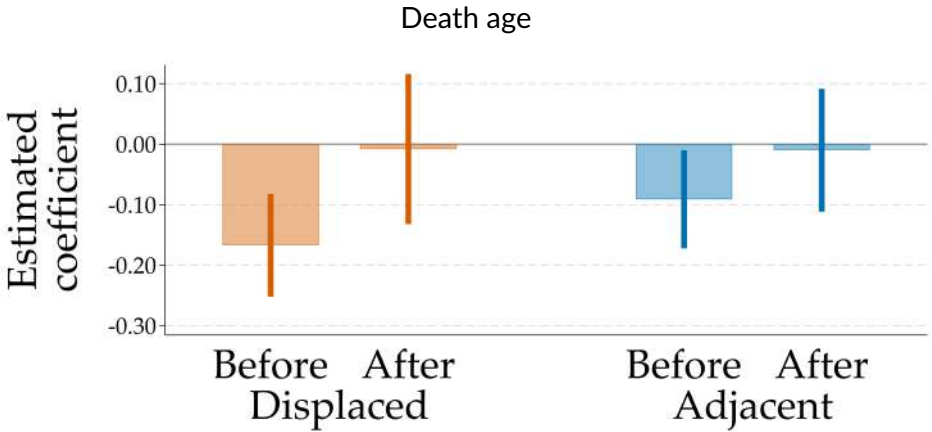
# Balance between those displaced before and after the law




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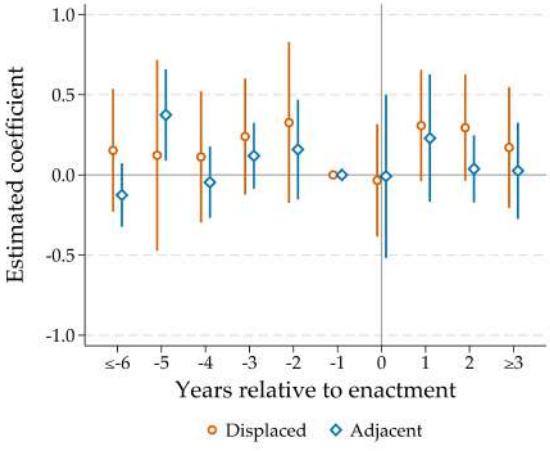
# Effects before and after the state enacted relocation assistance



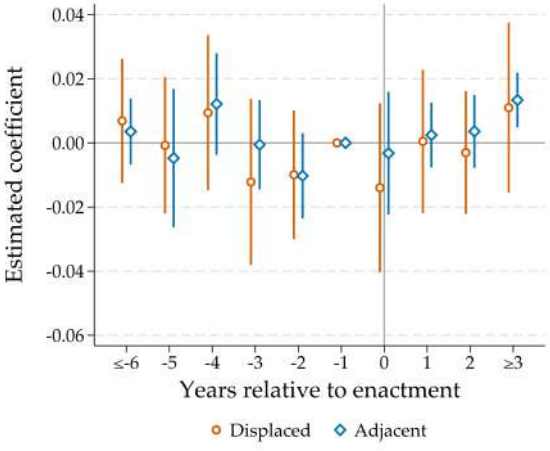

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# Event study around the enactment of relocation assistance

Death age



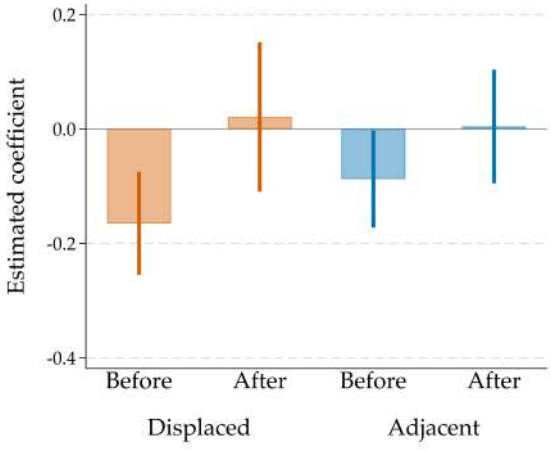

Same neighborhood



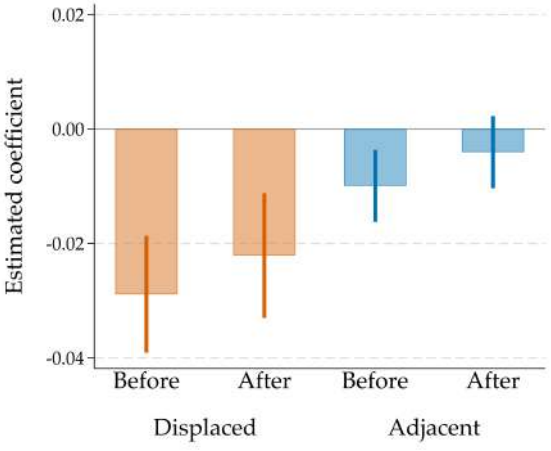

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# Results hold with state-by-construction-year fixed effects

Death age

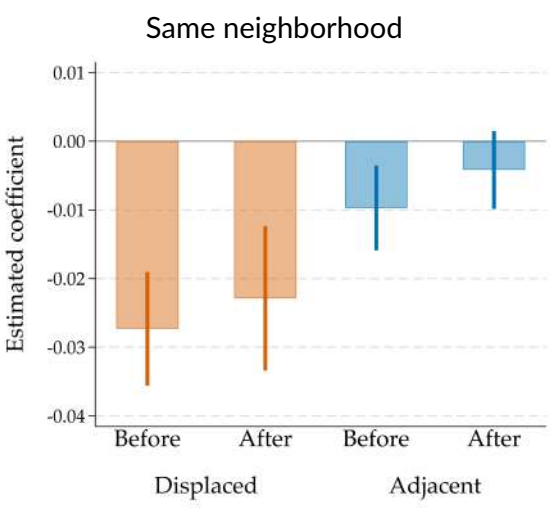


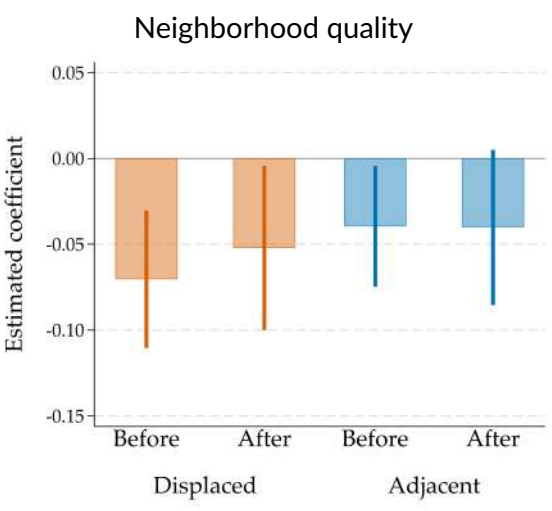

Same neighborhood




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# Assistance did not change where the displaced moved




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# Displaced movers were more likely to enter public housing

| | Public Housing | log Home Value | log Monthly Rent | log Annual Income |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Displaced by highway | 0.097[b] | -0.201[c] | -0.176 | -0.330[a] |
| | (0.046) | (0.107) | (0.118) | (0.115) |
| Previous homeowner | -0.001 | 0.213[a] | 0.057[b] | 0.291[a] |
| | (0.007) | (0.019) | (0.024) | (0.019) |
| Observations | 8,652 | 4,204 | 4,098 | 12,396 |
| R-squared (adjusted) | 0.026 | 0.079 | 0.229 | 0.055 |
| Mean dependent variable | 0.051 | 11.853 | 6.267 | 10.458 |

Movers in the last 12 months, 1973–1974 AHS. Robust SEs. [a] 1%, [b] 5%, [c] 10%.

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